Introduction to Econometrics
Week 1:
The classical regression model (Chapter 1.1)

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Based on Rob Alessie's lecture notes

Outline of the lecture

1. Overview of CLR
2. Assumptions of the CLR

Overview

- The center of the problem of econometrics lies in the way the data is gathered
- Depending on the estimating situation, different estimation method is appropriate
- Econometrics textbooks are usually organized as catalogues which are centred around the *Classical linear regression model* (CLR)
- In the most standard situation least squares estimator (OLS) is the optimal estimator

Overview

- CLR model consists of 4 assumptions
- By changing them in one way or another, different estimating situations are created
- In many of these situations OLS is no longer optimal
- Researcher needs to assess whether the 4 assumptions are met and if one or more are violated, then they turn to the catalogue (textbook) to see whether OLS is still optimal or what other options are available
- Often, it is not obvious whether assumptions are violated, so where possible textbooks include techniques useful in testing whether or not CLR model assumptions are violated

Overview

- CLR model posits a linear relationship between a **dependent variable** y and some **explanatory variables**
- Suppose we observe n values for those variables
- The **sample** or **data** is a collection of those n observations.
- Both the dependent and independent variables have to be treated as random variables, variables whose values are subject to chance.
- **Model:** set of restrictions on the joint distribution of the dependent and explanatory variables.

More Specific Overview

The classical regression model is a set of joint distributions satisfying 4 following assumptions

1. Linearity
2. Strict exogeneity
3. No perfect multicollinearity
4. Spherical error variance

Assumption 1.1: The linearity assumption

Assumption 1.1: the relationship between the dependent variable and the regressors is **linear**.

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + \varepsilon_i, \quad i = 1, \ldots, n \tag{1}$$

where

y_i:=**dependent variable**; other terms: **regressand** or **left hand side variable**

$(x_{i1}, \ldots, x_{iK})$:= K **explanatory variables**. Other terms: a) **regressors**, b) **right hand side variables**, c) **independent variables**.

$\beta_1, \ldots, \beta_K$: **regression coefficients**.

ε_i:= **disturbance term** or **error term** with certain properties to be specified below.

Remarks

- The part of the right-hand side involving the regressors

$$\beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_K x_{iK},$$

  is called the **regression** or the **regression function**.
- The disturbance term ε_i captures all influences on y_i left unexplained by the regressors (omitted variables, measurement error, inherent randomness in human behavior)
- In almost all cases, a regression model includes a constant term, i.e. $x_{i1} = 1$ for all i. In that case model (1) becomes

$$y_i = \beta_1 + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + \varepsilon_i, \quad i = 1, \ldots, n \tag{2}$$

  where β_1 is called the **intercept** (constant term). The other regression coefficients are called **slope coefficients**.

Remarks

- Slope coefficients represent the **marginal effect** of the regressors
- For example,

$$\beta_2 = \frac{\partial y_i}{\partial x_{i2}}$$

  i.e. β_2 represents the change in the dependent variable when the second regressor increases by one unit while other regressors are held constant (ceteris paribus)
- The linearity assumption implies that the marginal effect is constant and does not depend on the level of regressors

Example 1.1: wage equation

- Basic wage equation:

$$wage_i = \beta_1 + \beta_2 S_i + \beta_3 tenure_i + \beta_4 exper_i + \beta_5 exper_i^2 + \varepsilon_i \qquad (3)$$

  where $wage_i$:= hourly wage rate (let's say in US dollars); S_i:= years of schooling; $tenure_i$= no. of years on the current job; $exper_i$= experience in the labor force.

- When the equation has only one nonconstant regressor, it is called the **simple regression model**; if it has more than one nonconstant regressors (as in this example), it is called **multiple regression model**.

- Interpretation of the coefficients:

$$\frac{\partial wage_i}{\partial S_i} = \beta_2$$

  Suppose that $\beta_2 = 1$. **Interpretation of β_2:** A one year increase in S_i leads to a $1USD$ an increase in the hourly wage rate keeping other factors constant (ceteris paribus).

Example 1.1: wage equation

- Typically, the wage equation is specified in a **semi-log** form - this changes the interpretation!

$$\ln(wage_i) = \beta_1 + \beta_2 S_i + \beta_3 tenure_i + \beta_4 exper_i + \beta_5 exper_i^2 + \varepsilon_i \quad (4)$$

- Interpretation regression coefficients in a semi-log model:

$$\frac{\partial \ln(wage_i)}{\partial S_i} = \frac{\partial wage_i / wage_i}{\partial S_i} = \beta_2$$

Suppose that $\beta_2 = 0.05$.
Interpretation of β_2: A one year increase in S_i leads to 5 percent (β_2*100) increase in the hourly wage rate keeping other factors constant (ceteris paribus).

Example 1.1: wage equation

- Notice that in (4) we have experience and experience squared in the model
- This way we take into account that the returns to experience may decrease with $exper_i$ ($\beta_5 < 0$)

$$\frac{\partial \ln(wage_i)}{\partial exper_i} = \beta_4 + 2\beta_5 exper_i,$$

- That's one way to account for a non-linear relationship between the dependent and independent variables in the CRM framework (model is still linear as it is linear in parameters) $\rightarrow$ assumption 1.1. is satisfied.
- Also, note that ε_i captures the impact of variables not included in the regression, for instance "innate ability", "motivation"

CLR Model in Matrix notation

- Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{iK})'$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)'$ be $(K \times 1)$ vectors, then linearity assumption 1.1 (model (1)) can be rewritten as follows:

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \ldots, n \tag{5}$$

because by the definition of vector inner product:

$$\mathbf{x}_i' \boldsymbol{\beta} = \beta_1 x_{i1} + \ldots + \beta_K x_{iK}$$

Stacking the n observations gives:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{6}$$

where $\mathbf{y} = (y_1, ..., y_n)'$; $\boldsymbol{\varepsilon} = (\varepsilon_1, ..., \varepsilon_n)'$ are $(n \times 1)$ vectors and $\mathbf{X}$ a $(n \times K)$-matrix defined as follows:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \ldots \\ \mathbf{x}_n' \end{pmatrix} = \begin{pmatrix} x_{11} \ldots x_{1K} \\ x_{21} \ldots x_{2K} \\ \vdots \ldots \vdots \\ x_{n1} \ldots x_{nK} \end{pmatrix}$$

Assumption 1.2: The strict exogeneity assumption

$$E(\varepsilon_i|\mathbf{X}) = 0, \quad i = 1, \ldots, n \tag{7}$$

- (Important!) the expectation is conditional on the regressors for *all* observations.
- This becomes more clear if we rewrite the strict exogeneity as follows:

$$E(\varepsilon_i|\mathbf{x}_1, \ldots, \mathbf{x}_n) = 0, \quad i = 1, \ldots, n \tag{8}$$

- To state the assumption differently, take for any given observation i, the **joint distribution** of the $nK + 1$ random variables, $f(\varepsilon_i, \mathbf{x}_1, \ldots, \mathbf{x}_n)$, and consider the **conditional distribution**, $f(\varepsilon_i|\mathbf{x}_1, \ldots \mathbf{x}_n)$.
  - The conditional mean $E(\varepsilon_i|\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is in general a nonlinear function of $\mathbf{x}_1, \ldots, \mathbf{x}_n$.
  - The strict exogeneity assumption says that this function is a constant of value zero.

Assumption 1.2: The strict exogeneity assumption

- Assuming this constant to be zero is not restrictive if the regressors include a constant.
- To see this, suppose that $E(\varepsilon_i|\mathbf{X}) = \mu$ and $x_{i1} = 1$. Then equation (1) can be rewritten as:

$$\begin{aligned} y_i &= \beta_1 + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + \varepsilon_i \\ &= (\beta_1 + \mu) + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + (\varepsilon_i - \mu) \\ &= \beta_1^* + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + \varepsilon_i^* \end{aligned} \tag{9}$$

Notice that $E(\varepsilon_i^*|\mathbf{X}) = 0$.

In this case the estimate of the intercept is biased though.

Implications of strict exogeneity

First, we need to be reminded of the following 8 facts:

- Fact 1: **Law of total expectations**:

$$E[E(y|\mathbf{x})] = E(y)$$

- Fact 2: Two random variables x and y are **orthogonal** to each other if

$$E(xy) = 0$$

- Fact 3: **Law of iterated expectations**:

$$E[E(y|\mathbf{x}, \mathbf{z})|\mathbf{x}] = E(y|\mathbf{x})$$

- Fact 4: **Linearity of conditional expectations**:

$$E(f(\mathbf{x})y|\mathbf{x}) = f(\mathbf{x})E(y|\mathbf{x})$$

Implications of strict exogeneity

- Fact 5: The **Covariance** of two random variables x and y:

$$Cov(x, y) = E((x - E(x))(y - E(y)) = E(xy) - E(x)E(y) \quad (10)$$

- Fact 6: Consider the covariance between x and y. Suppose that $E(y) = 0$. Then equation (10) implies that

$$Cov(x, y) = E(xy) \quad (11)$$

- Fact 7: **Correlation coefficient** between two random variables:

$$\rho(x, y) = \frac{Cov(x, y)}{sd(x)sd(y)}$$

- Fact 8: Two random variables x and y are **uncorrelated** to each other if

$$\rho(x, y) = Cov(x, y) = 0$$

Implications of strict exogeneity

Having established facts 1-8 we can now talk about four important implications of the strict exogeneity assumption, $E(\varepsilon_i|\mathbf{X}) = 0$:

1. **The unconditional mean of the error term is zero**, i.e. $E(\varepsilon_i) = 0$
Proof: According to the law of total expectations:

$$E(\varepsilon_i) = E(E(\varepsilon_i|\mathbf{X})) = E(0) = 0 \tag{12}$$

q.e.d.

Implications of strict exogeneity

2. **Under strict exogeneity, the regressors are orthogonal to the error term for all observations**, i.e.,

$$E(x_{jk}\varepsilon_i) = 0, \quad i, j = 1, \ldots, n; k = 1, \ldots, K \tag{13}$$

(these are so called **othogonality conditions**) **Proof:**

$$\begin{aligned} E(x_{jk}\varepsilon_i) = E[E(x_{jk}\varepsilon_i|x_{jk})] = & \text{ (by Law of total expectations)} \\ E[x_{jk}E(\varepsilon_i|x_{jk})] = & \text{ (by linearity of conditional expectations)} \\ E[x_{jk}0] = 0 & \end{aligned} \tag{14}$$

The last line of equation (14) follows from the following:

$$E(\varepsilon_i|x_{jk}) = E[E(\varepsilon_i|\mathbf{X})|x_{jk}] = E[0|x_{jk}] = 0 \tag{15}$$

In the last step, since x_{jk} is an element of the data matrix $\mathbf{X}$, we can apply the law of iterated expectations.

Implications of strict exogeneity

3. **Because the mean of the error term is zero, the orthogonality conditions (13) are equivalent to zero-correlation conditions**. This is because

$$Cov(\varepsilon_i, x_{jk}) = E(\varepsilon_i x_{jk}) - E(\varepsilon_i)E(x_{jk}) = E(\varepsilon_i x_{jk}) = 0 \qquad (16)$$

The second equality follows from $E(\varepsilon_i) = 0$ (cf. equation (12), the last equality from orthogonality conditions (13).

In particular, for $i = j$, $Cov(x_{ik}, \varepsilon_i) = 0$ - regressors are uncorrelated with the error term. Consider the following model:

$$\ln(wage_i) = \beta_1 + \beta_2 s_i + \beta_3 tenure_i + \beta_4 exper_i + \beta_5 exper_i^2 + \varepsilon_i$$

where s_i denotes years of schooling

What does implication (3) mean in this model?

Implications of strict exogeneity

4. **Together with the linearity assumption 1.1, the strict exogeneity assumption also implies that**

$$E(y_i|\mathbf{X}) = E(\mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i|X) = \mathbf{x}_i'\boldsymbol{\beta} + E(\varepsilon_i|\mathbf{X}) = \mathbf{x}_i'\boldsymbol{\beta} \tag{17}$$

$\Rightarrow$ If we know $\boldsymbol{\beta}$, we can predict y_i!

Assumption 1.3

Assumption 1.3 (no perfect multicollinearity): The column rank of the $n \times K$ data matrix, $\mathbf{X}$, is K with probability 1.

Comments on no perfect multicollinearity

- The rank of a matrix equals the number of linearly independent columns of the matrix.
- The assumption of full column rank says that none of the K columns of the data matrix $\mathbf{X}$ can be expressed as a linear combination of the other columns of $\mathbf{X}$.
- Since the K columns cannot be linearly independent if their dimension is less than K, the assumption implies that $n \geq K$, i.e., there must be at least as many observations as there are regressors.
- The regressors are said to be **perfectly multicollinear** if assumption 1.3 is not satisfied.

Assumption 1.3: Wage equation example

- Suppose that we have cross-section data available for the year 1996
- Consider the following model

$$log(wage_i) = \beta_1 + \beta_2 s_i + \beta_3 age_i + \beta_4 yob_i + \varepsilon_i \tag{18}$$

where

age_i= age of individual i (measured in years)
yob_i= year of birth of individual i

- Obviously, $yob_i = 1996 - age_i$ for all i.
- Since model (18) includes a constant term ($x_{i1} = 1$), the regressor yob_i is a linear combination of the constant term and the the variable age. In other words, we face a problem of perfect multicollinearity.
- Given the identity $yob_i = 1996 - age_i$, we can rewrite equation (18) as follows:

$$log(wage_i) = (\beta_1 + 1996\beta_4) + \beta_2 s_i + (\beta_3 - \beta_4) age_i + \varepsilon_i \tag{19}$$

which shows that only the difference $\beta_3 - \beta_4$ but not β_3 and β_4 separately, can be estimated.

Assumption 1.4

Assumption 1.4 (spherical error variance):

(homoskedasticity)

$$E(\varepsilon_i^2|\mathbf{X}) = \sigma^2 > 0, \quad i = 1, \ldots, n \tag{20}$$

(no correlation between observations)

$$E(\varepsilon_i \varepsilon_j|\mathbf{X}) = 0, \quad i, j = 1, \ldots, n; i \neq j \tag{21}$$

- The homoskedasticity assumption (20) says that the conditional second moment of ε_i, which in general is a nonlinear function of $\mathbf{X}$, is a constant.
- Since the (i, j) element of the $n \times n$ matrix $\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'$ is $\varepsilon_i \varepsilon_j$, Assumption 1.4 can be written compactly as

$$E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|\mathbf{X}) = \sigma^2 \mathbf{I}_n \tag{22}$$

  where $\mathbf{I}_n$ denotes the identity matrix of order n
- It is called the **spherical** error variance assumption because the $n \times n$ matrix of second moments (which are also variances and covariances) is proportional to the identity matrix $\mathbf{I}_n$.

Assumption 1.4

Comments on spherical error variance:

- Thanks to strict exogeneity, the homoskedasticity assumption can be stated equivalently in more familiar terms.
- Consider the conditional variance

$$Var(\varepsilon_i|\mathbf{X}) = E(\varepsilon_i^2|\mathbf{X}) - [E(\varepsilon_i|\mathbf{X})]^2 = E(\varepsilon_i^2|\mathbf{X}) \tag{23}$$

  where in the second equality we use the assumption of strict exogeneity: $E(\varepsilon_i|\mathbf{X}) = 0$
  So we can restate assumption (20) as follows:

$$Var(\varepsilon_i|\mathbf{X}) = \sigma^2 > 0, \quad i = 1, \ldots, n \tag{24}$$

Assumption 1.4

- Similarly, assumption (21; no correlation between observations) is equivalent to the requirement that

$$Cov(\varepsilon_i, \varepsilon_j | \mathbf{X}) = 0 \quad i, j = 1, \ldots, n; i \neq j \tag{25}$$

To see this notice that by definition

$$Cov(\varepsilon_i, \varepsilon_j | \mathbf{X}) = E(\varepsilon_i \varepsilon_j | \mathbf{X}) - E(\varepsilon_i | \mathbf{X}) E(\varepsilon_j | \mathbf{X}) \tag{26}$$

Due to the assumption of strict exogeneity, the second term of the rhs of equation (26) cancels out.

The classical regression model for random samples

- In this course, we will assume that we work with **random samples**
- We have a random sample of size n, i.e.

$$\{(x_{1i}, \ldots, x_{Ki}, y_i) : i = 1, \ldots, n\}$$

  are independent, identically distributed (I.I.D.) random variables across observations.
- Since:
  a by the linearity assumption 1.1 ε_i is a function of $(y_i, \mathbf{x}_i)$ and
  b $(y_i, \mathbf{x}_i)$ is independent of $(y_j, \mathbf{x}_j)$ for $i \neq j$,

  $(\varepsilon_i, \mathbf{x}_i)$ is independent of $\mathbf{x}_j$ for $i \neq j$. So:

$$\begin{aligned} E(\varepsilon_i|\mathbf{X}) &= E(\varepsilon_i|\mathbf{x}_1, \ldots, \mathbf{x}_{i-1}, \mathbf{x}_i, \mathbf{x}_{i+1}, \ldots, \mathbf{x}_n) = E(\varepsilon_i|\mathbf{x}_i) \\ E(\varepsilon_i^2|\mathbf{X}) &= E(\varepsilon_i^2|\mathbf{x}_i) \qquad (27) \\ \text{and } E(\varepsilon_i\varepsilon_j|\mathbf{X}) &= E(\varepsilon_i|\mathbf{x}_i)E(\varepsilon_j|\mathbf{x}_j) \qquad (28) \end{aligned}$$

- Therefore assumptions 1.2 (strict exogeneity) and 1.4 reduce to

$$\begin{aligned} \text{Assumption 1.2 } E(\varepsilon_i|\mathbf{x}_i) &= 0 \qquad (29) \\ \text{Assumption 1.4 } E(\varepsilon_i^2|\mathbf{x}_i) &= \sigma^2 \quad i = 1, \ldots, n \qquad (30) \end{aligned}$$

"Fixed" regressors

- Up to now, we have treated the regressors as random.
- This is in contrast to the treatment in most older textbooks, where $\mathbf{X}$ is assumed to be "fixed" or deterministic.
- The assumption that x_i is fixed implies that it is possible to redraw the sample with the same explanatory variable values
- If $\mathbf{X}$ is fixed, then there is no need to distinguish between the **conditional distribution** of the error term, $f(\varepsilon_i|\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and the marginal distribution, $f(\varepsilon_i)$
- In other words, assumptions 1.2 and 1.4 can be written as

$$\begin{aligned} \text{Assumption 1.2} \ : E(\varepsilon_i) &= 0 && (31) \\ \text{Assumption 1.4} \ : E(\varepsilon_i^2) &= \sigma^2 \quad i = 1, \ldots, n && (32) \\ E(\varepsilon_i \varepsilon_j) &= 0 && (33) \end{aligned}$$